ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE H

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): fitness for purpose and a measurable specification, the block diagram and closed-loop control of an automated school greenhouse, hysteresis and a ladder diagram, a work breakdown structure, sizing the solar panel and battery, budgets and drawings, calibrating sensors, a spider diagram for handover, a snag list, and maintenance schedules, data logging and electrical faults.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw sketches, tables, diagrams and circuits neatly. Use only a black or blue pen.

> **PROJECT.** A rural school grows vegetables in a greenhouse, but the plants die during school holidays because nobody waters them. The school wants an automated greenhouse that runs on solar power. Soil-moisture sensors switch an irrigation pump on when the soil is dry, a thermostat switches an extractor fan on when the greenhouse is too hot, a float switch protects the pump when the water tank is low, and a data logger records the conditions.

QUESTION 1: THE DESIGN

1.1 Explain what is meant by fitness for purpose in a design. (2)

1.2 Write FOUR items of the design specification for the greenhouse system. Give a measurable value for EACH item. (4)

1.3 Explain why the use of water and energy must be considered in this design. (2)

1.4 Give TWO ways in which design ideas can be presented to the school. (2)

[10]

QUESTION 2: AUTOMATING THE GREENHOUSE

2.1 Draw a block diagram of the greenhouse control system. Show the inputs, the controller, the outputs and the power supply. (6)

2.2 Explain how the soil-moisture sensor, the controller and the pump form a closed-loop control system. (4)

2.3 The pump switches on when the soil moisture falls to 30% and switches off only when it reaches 60%. Explain what this hysteresis is and why it is used. (4)

2.4 Draw a PLC ladder diagram with these inputs and outputs: AUTO switch I0.0; soil-dry sensor I0.1 (ON when the soil is dry); tank float switch I0.2 (ON while there is enough water); thermostat I0.3 (ON when the greenhouse is too hot); pump Q0.0; fan Q0.1; "tank low" warning lamp Q0.2. The pump may run only in AUTO when the soil is dry and there is enough water. The fan runs in AUTO when it is too hot. The lamp is on whenever the tank is low. (8)

2.5 Explain why PVC or polyethylene pipe is used for the irrigation lines. (3)

[25]

QUESTION 3: PROJECT MANAGEMENT

3.1 Explain what a work breakdown structure (WBS) is, and draw one for the greenhouse project with at least SIX tasks. (8)

3.2 The system uses a 60 W pump for 3 hours a day, a 40 W fan for 5 hours a day and a 5 W controller and data logger for 24 hours a day. The area gets 5 hours of full sun a day. Calculate the: (8)

3.2.1 total energy used per day in watt-hours (3)

3.2.2 solar panel power needed if 25% extra is added for losses (3)

3.2.3 battery capacity in ampere-hours for a 12 V battery to run the system for one day without sun (2)

3.3 Explain the difference between the budget and the actual cost of a project, and why the difference must be tracked. (4)

3.4 Explain how EACH of the following drawings is used during installation: (6)

3.4.1 Circuit diagram (2)

3.4.2 Wiring diagram (2)

3.4.3 Layout drawing of the greenhouse (2)

3.5 Name FOUR items of personal protective equipment (PPE) for the installation team and give the reason for EACH. (4)

[30]

QUESTION 4: TESTING AND COMMISSIONING

4.1 Explain how the soil-moisture sensor is calibrated during commissioning. (3)

4.2 Draw a spider diagram that shows FIVE things that must be tested before the greenhouse system is handed over. (5)

4.3 Explain what a snag list (punch list) is. (2)

4.4 During testing the pump runs, but no water reaches the plants. Give THREE possible causes. (3)

4.5 Explain why the teachers and the caretaker must be trained to use the system. (2)

[15]

QUESTION 5: MAINTENANCE

5.1 Draw up a maintenance schedule for the greenhouse system with SIX tasks: two weekly, two monthly and two yearly. (6)

5.2 Explain how the solar panels are cleaned and checked safely. (3)

5.3 Explain what the data logger records and how this information helps with maintenance. (4)

5.4 Explain the difference between a short-circuit fault and an earth fault. (4)

5.5 Give THREE reasons why maintenance records must be kept. (3)

[20]

TOTAL: 100

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END OF QUESTION PAPER